

Physics - Question 81

Original paper page 30; original question and option regions

81. Math the list -I with list -II

List -I

A) Boltzman constant

B) Coefficient of viscosity

C) Water equivalent

D) Coefficient of thermal conductivity

The correct match in the following is

1) A-III;B-I;C-II;D-IV

3) A-IV;B-II;C-I;D-III

Key: 3**Sol:** . A – IV, B – II, C – I, D – III

$$PV = NKT$$

$$K = \frac{PV}{NT} = \frac{M^1 L^2 T^2}{K^1} = M^1 L^2 T^{-2} K^{-1}$$

$$\eta = \frac{\tan \text{ gential stress}}{\text{vel gradient}} = \frac{M^1 L^{-1} T^{-2}}{T^{-1}} = M^1 L^{-1} T^{-1}$$

$$\text{Water equivalent } W = \text{mass} = M^1$$

$$\text{Coefficient of thermal conductivity } K = \frac{Qd}{A t \Delta \theta}$$

$$= M^1 L^1 T^{-3} K^{-1}$$

List -II

I) ML^0T^0 II) $ML^{-1}T^{-1}$ III) $MLT^{-3}K^{-1}$ IV) $ML^2T^{-2}K^{-1}$

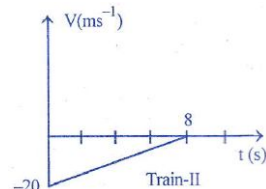
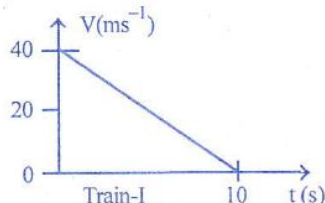
2) A-III;B-II;C-I;D-IV

4) A-IV;B-I;C-II;D-III

Physics - Question 82

Original paper page 30; original question and option regions

82. Two trains, which are moving along different tracks in opposite directions are put on the same track by mistake. On noticing the mistake, when the trains are 300 m apart the drivers start slowing down the trains. The graphs given below show decreases in their velocities as function of time. The separation between the trains when both have stopped is



1) 120 m

2) 20 m

3) 60 m

4) 280 m

Key: 2**Sol:** Area under V-t graph = displacement

Distance travelled by first train

$$= \frac{1}{2} \times 40 \times 10 = 200 \text{ m}$$

Distance travelled by second train

$$= \frac{1}{2} \times 20 \times 8 = 80 \text{ m}$$

The separation between the trains when they have stopped

$$= 300 \text{ m} - 200 \text{ m} - 80 \text{ m} = 20 \text{ m}$$

Physics - Question 83

Original paper page 31; original question and option regions

83. A point object moves along an arc of a circle of radius 'R'. Its velocity depends upon the distance covered 'S' as $V = K\sqrt{S}$ where 'K' is a constant. If ' θ ' is the angle between the total acceleration and tangential acceleration, then

$$1) \tan \theta = \sqrt{\frac{S}{R}} \quad 2) \tan \theta = \sqrt{\frac{S}{2R}} \quad 3) \tan \theta = \frac{S}{2R} \quad 4) \tan \theta = \frac{2S}{R}$$

Key : 4

Sol : $V = K\sqrt{S}$

$$\tan \theta = \frac{a_r}{a_t} = \frac{V^2}{R \left(\frac{dv}{dt} \right)}$$

$$a_t = \frac{dv}{dt} = \frac{dv}{ds} \times \frac{ds}{dt} = \frac{v dv}{ds} = k\sqrt{s} \times k \times \frac{1}{2\sqrt{s}} = \frac{K^2}{2}$$

$$\therefore \tan \theta = \frac{v^2}{R \times \frac{k^2}{2}} = \frac{2v^2}{k^2 R}$$

$$\tan \theta = \frac{2}{k^2 R} \times k^2 s = \frac{2s}{R}$$

Physics - Question 84

Original paper page 31; original question and option regions

84. A body projected from the ground reaches a point 'X' in its path after 3 seconds and from there it reaches the ground after further 6 seconds. The vertical distance of the point 'X' from the ground is (acceleration due to gravity $= 10 \text{ms}^{-2}$)

1) 30 m

2) 60 m

3) 80 m

4) 90 m

Key : 4

Sol :

Time of flight = $T = 9\text{s}$

$$\frac{2u}{g} = T = 9\text{s}$$

$$u = \frac{90}{2} = 45 \text{ m/s}$$

$$h = ut - \frac{1}{2}gt^2$$

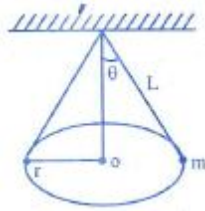
$$h = (45)(3) - \frac{1}{2} \times 10 \times 9$$

$$= 90\text{m}$$

Physics - Question 85

Original paper page 31; original question and option regions

85. A particle of mass 'm' is suspended from a ceiling through a string of length 'L'. If the particle moves in a horizontal circle of radius 'r' as shown in the figure, then the speed of the particle is



1) $r\sqrt{\frac{g}{\sqrt{L^2 - r^2}}}$

2) $g\sqrt{\frac{r}{\sqrt{L^2 - r^2}}}$

3) $r\sqrt{\frac{g}{L^2 - r^2}}$

4) $g\sqrt{\frac{r}{L^2 - r^2}}$

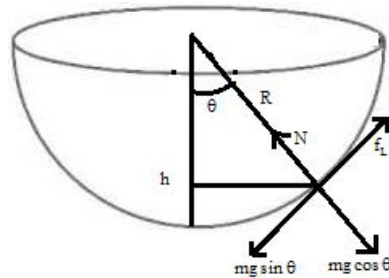
Key : 1

Physics - Question 86

Original paper page 32; original question and option regions

86. A particle is placed at rest inside a hollow hemisphere of radius 'R'. The coefficient of friction between the particle and the hemisphere is $\mu = \frac{1}{\sqrt{3}}$. The maximum height upto which the particle can remain stationary is

- 1) $\frac{R}{2}$ 2) $\left(1 - \frac{\sqrt{3}}{2}\right)R$ 3) $\frac{\sqrt{3}}{2}R$ 4) $\frac{3R}{8}$



Key : 2

Sol: At maximum height

$$f_L = mg \sin \theta = \mu N = \mu mg \cos \theta$$

$$\tan \theta = \mu$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = 30^\circ$$

$$\cos \theta = \frac{R - h}{R}$$

$$\frac{\sqrt{3}}{2} = \frac{R - h}{R}$$

$$h = R \left(\frac{2 - \sqrt{3}}{2} \right)$$

Physics - Question 87

Original paper page 33; original question and option regions

87. A 1 kg ball moving with a speed of 6ms^{-1} collides head-on with a 0.5 kg ball moving in the opposite direction with a speed of 9ms^{-1} . If the co-efficient of restitution is $\frac{1}{3}$, the energy lost in the collision is

1) 303.4 J

2) 66.7 J

3) 33.3 J

4) 67.8 J

Key : 3

Sol :

$$\Delta\text{KE} = \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) (u_1 - u_2)^2 (1 - e^2)$$

$$\Delta\text{KE} = \frac{1}{2} \times \left(\frac{1}{\frac{2}{3/2}} \right) \times (15)^2 \times \left(1 - \frac{1}{9} \right)$$

$$= 33.3\text{J}$$

Physics - Question 88

Original paper page 33; original question and option regions

88. A ball is thrown vertically down from a height of 40 m from the ground with an initial velocity 'v'. The ball hits the ground, loses $\frac{1}{3}$ of its total mechanical energy and rebounds back to the same height. If the acceleration due to gravity is 10 ms^{-2} , the value of 'v' is

1) 5 ms^{-1}

2) 10 ms^{-1}

3) 15 ms^{-1}

4) 20 ms^{-1}

Key : 4

$$\text{Sol: } \frac{2}{3} \left(\frac{1}{2} mv^2 + mgh \right) = mgh$$

$$\frac{v^2}{2} + gh = \frac{3}{2} gh$$

$$v = \sqrt{gh}$$

$$= \sqrt{10 \times 40} = 20 \text{ m/s}$$

Physics - Question 89

Original paper page 33; original question and option regions

- 89. Three identical uniform thin metal rods form the three sides of an equilateral triangle. If the moment of inertia of the system of these three rods about an axis passing through the centroid of the triangle and perpendicular to the plane of the triangle is 'n' times the moment of inertia of one rod separately about an axis passing through the centre of the rod and perpendicular to its length, the value of 'n' is**

1) 3

2) 6

3) 9

4) 12

Key : 2

Sol : $I_1 = nI_0$

$$3\left(\frac{ml^2}{12} + m(r^2)\right) = n\left(\frac{ml^2}{12}\right)$$

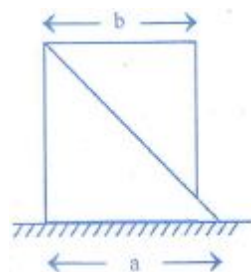
$$3\left(\frac{ml^2}{12} + m\frac{l^2}{12}\right) = n\left(\frac{ml^2}{12}\right)$$

$$n = 6$$

Physics - Question 90

Original paper page 34; original question and option regions

90. Two smooth and similar right angled prisms are arranged on a smooth horizontal plane as shown in the figure. The lower prism has a mass '3' times the upper prism. The prisms are held in an initial position as shown and are then released. As the upper prism touches the horizontal plane, the distance moved by the lower prism is



- 1) $a - b$ 2) $\frac{a - b}{3}$ 3) $\frac{b - a}{2}$ 4) $\frac{a - b}{4}$

Key : 4

Sol : momentum of upper prism in the right direction = momentum of lower prism in the left direction

$$\frac{m(a - b - x)}{t}$$

$$= 3m \frac{x}{t}$$

$$(a - b) = 4x$$

$$x = \frac{a - b}{4}$$

Physics - Question 91

Original paper page 34; original question and option regions

91. A Particle is executing simple harmonic motion with an amplitude of 2 m. The difference in the magnitudes of its maximum acceleration and maximum velocity is 4. The time period of its oscillation and its velocity when it is 1 m away from the mean position are respectively.

- 1) $2\text{s}, 2\sqrt{3}\text{ms}^{-1}$ 2) $\frac{7}{22}\text{s}, 4\sqrt{3}\text{ms}^{-1}$ 3) $\frac{22}{7}\text{s}, 2\sqrt{3}\text{ms}^{-1}$ 4) $\frac{44}{7}\text{s}, 4\sqrt{3}\text{ms}^{-1}$

Key ; 3

$$\text{Sol : } a_{\max} - v_{\max} = 4$$

$$\omega^2 A - \omega A = 4$$

$$\omega^2 \times 2 - \omega \times 2 = 4$$

$$\Rightarrow \omega = 2$$

$$\frac{2\pi}{T} = 2$$

$$T = \pi = \frac{22}{7}\text{sec}$$

$$v = \omega \sqrt{A^2 - y^2} = \frac{2\pi}{T} \sqrt{2^2 - 1^2}$$

$$v = 2\sqrt{3}\text{m/s}$$

Physics - Question 92

Original paper page 35; original question and option regions

92. Two bodies of masses 'm' and '9m' are placed at a distance 'r'. The gravitational potential at a point on the line joining them, where gravitational field is zero, is (G is universal gravitational constant)

1) $\frac{-14Gm}{r}$

2) $\frac{-16Gm}{r}$

3) $\frac{-12Gm}{r}$

4) $\frac{-8Gm}{r}$

Key : 2

Sol : Let x be the distance from m where gravitational field is zero

$$\Rightarrow \frac{Gm}{x^2} = \frac{G(9m)}{(r-x)^2}$$

$$x = \frac{r}{4}$$

$$\text{Potential} = -\frac{Gm}{x} - \frac{G(9m)}{r-x}$$

$$= \frac{-Gm}{\left(\frac{r}{4}\right)} - \frac{9Gm}{\left(\frac{3r}{4}\right)}$$

$$= \frac{-4Gm}{r} - \frac{12Gm}{r}$$

$$= -\frac{16Gm}{r}$$

Physics - Question 93

Original paper page 35; original question and option regions

93. When a load of 80 N is suspended from a string, its length is 101 mm. If a load of 100 N is suspended, its length is 102 mm. If a load of 160 N is suspended from it, then length of the string is (Assume the area of cross-section unchanged)

1) 15.5 cm

2) 13.5 cm

3) 16.5 cm

4) 10.5 cm

Key : 4

Sol : Load $\propto$ extension

Let initial length of string = l_0

$$80 = k(101 - l_0)$$

$$100 = k(102 - l_0)$$

$$l_0 = 97 \text{ mm}$$

$$160 = k(l - 97)$$

$$80 = k(101 - 97)$$

$$l = 105 \text{ mm} = 10.5 \text{ cm}$$

Physics - Question 94

Original paper page 35; original question and option regions

94. A sphere of material of relative density 8 has a concentric spherical cavity and just sinks in water. If the radius of the sphere is 2 cm, then the volume of the cavity is

1) $\frac{76}{3} \text{ cm}^3$

2) $\frac{79}{3} \text{ cm}^3$

3) $\frac{82}{3} \text{ cm}^3$

4) $\frac{88}{3} \text{ cm}^3$

Key : 4

Physics - Question 95

Original paper page 36; original question and option regions

- 95. A hunter fired a metallic bullet of mass 'm' kg from a gun towards an obstacle and it just melts when it is stopped by the obstacle. The initial temperature of the bullet is 300 K . If $\frac{1^{\text{th}}}{4}$ of heat is absorbed by the obstacle, then the minimum velocity of the bullet is**
[Melting point of bullet = 600 K,
Specific heat of bullet = $0.03 \text{ cal g}^{-1} \text{ }^{\circ}\text{C}^{-1}$,
Latent heat of fusion of bullet = 6 cal g^{-1}]

1) 410 ms^{-1}

2) 410 ms^{-1}

3) 460 ms^{-1}

4) 310 ms^{-1}

Key : 1

$$\text{Sol : } \frac{3}{4} \times \Delta \text{KE} = ms\Delta t + mL$$

$$\frac{3}{4} \times \frac{1}{2} mv^2 = ms\Delta t + mL$$

$$V = \sqrt{\frac{8}{3}s\Delta t + \frac{8}{3}L}$$

$$V = \sqrt{\frac{8}{3} \times 0.03 \times 4200 \times 300 + \frac{8}{3} \times 6 \times 4200} = 410 \text{ ms}^{-1}$$

Physics - Question 96

Original paper page 36; original question and option regions

96. 'M' kg of water at 't' $^{\circ}\text{C}$ is divided into two parts so that one part of mass 'm' kg when converted into ice at 0°C would release enough heat to vapourise the other part, then $\frac{m}{M}$ is equal to

[Specific heat of water = $1 \text{ cal g}^{-1} ^{\circ}\text{C}^{-1}$

Latent heat of fusion of ice = 80 cal g^{-1}

Latent heat of steam = 540 cal g^{-1}]

1) $640 - t$

2) $\frac{720 - t}{640}$

3) $\frac{640 + t}{720}$

4) $\frac{640 - t}{720}$

Physics - Question 97

Original paper page 37; original question and option regions

97. A diatomic gas ($\gamma = 1.4$) does 300 J work when it is expanded isobarically. The heat given to the gas in this process is

1) 1050 J

2) 950 J

3) 600 J

4) 550 J

Key : 1

Sol : For isobaric process

$$Q : \Delta U : W = n C_p \Delta T : n C_v \Delta T : n R \Delta T$$

$$\Rightarrow \frac{W}{Q} = \frac{2}{7} \quad \frac{W}{Q} = \frac{R}{C_p} = \frac{\gamma - 1}{\gamma} = \frac{1.4 - 1}{1.4} = \frac{2}{7}$$

$$Q = \frac{7}{2} W = \frac{7}{2} \times 300 = 1050 \text{ J}$$

Physics - Question 98

Original paper page 37; original question and option regions

98. When the absolute temperature of the source of a Carnot heat engine is increased by 25% its efficiency increases by 80% .The new efficiency of the engine is

1) 12%

2) 24%

3) 48%

4) 36%

Key :4

$$\text{Sol : } \eta = 1 - \frac{T_2}{T_1}$$

$$1.8\eta = 1 - \frac{4T_2}{5T_1}$$

$$1.8\eta = 1 - \frac{4}{5}(1 - \eta)$$

$$1.8\eta = 1 - \frac{4}{5} + \frac{4}{5}\eta$$

$$1.8\eta - 0.8\eta = \frac{1}{5}$$

$$\eta = \frac{1}{5}$$

$$\therefore \text{ new efficiency} = 1.8 \times \frac{1}{5}$$

$$= 36\%$$

Physics - Question 99

Original paper page 37; original question and option regions

99. A cylinder of fixed capacity 67.2 litres contains helium gas at STP. The amount of heat needed to rise the temperature of the gas in the cylinder by 20°C is $(R = 8.31\text{Jmol}^{-1}\text{K}^{-1})$

1) 748 J

2) 374 J

3) 1000 J

4) 500 J

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Physics - Question 100

Original paper page 38; original question and option regions

100. For a certain organ pipe, three successive resonance frequencies are observed at 425, 595 and 765 Hz, respectively. The length of the pipe is (speed of sound in air = 340 ms^{-1})

1) 0.5 m

2) 1 m

3) 1.5 m

4) 2 m

Key : 2

Sol : frequencies are in the ratio

5 : 7 : 9

$\Rightarrow$ closed pipe

Fifth harmonic frequency of the closed pipe $\frac{5v}{4l} = 425$

$l = 1\text{m}$

Physics - Question 101

Original paper page 38; original question and option regions

101. A student holds a tuning fork oscillating at 170 Hz. He walks towards a wall at a constant speed of 2 ms^{-1} . The beat frequency observed by the student between the tuning fork and its echo is (Velocity of sound = 342 ms^{-1})

1) 2.5 Hz

2) 3 Hz

3) 1 Hz

4) 2 Hz

Key : 4

$$\text{Sol : } n_1 = n \left(\frac{v + v_0}{v - v_s} \right)$$

$$\text{here } v_s = v_0 = 2 \text{ m/s}$$

$$n_1 = 172 \text{ Hz}$$

$$\text{beat frequency } \Delta n = n_1 - n_2$$

$$= 172 - 170$$

$$= 2 \text{ Hz}$$

Physics - Question 102

Original paper page 38; original question and option regions

102. An infinitely long rod lies along the axis of a concave mirror of focal length 'f'. The nearer end of the rod is at a distance u, ($u > f$) from the mirror. It's image will have a length

1) $\frac{uf}{u+f}$

2) $\frac{uf}{u-f}$

3) $\frac{f^2}{u+f}$

4) $\frac{f^2}{u-f}$

Key : 4

Sol : Image of near end :

$$\frac{1}{f} = \frac{1}{v_1} + \frac{1}{u}$$

$$\frac{1}{-f} = \frac{1}{v_1} - \frac{1}{u}$$

Physics - Question 103

Original paper page 39; original question and option regions

103. In Young's double slit experiment, red light of wavelength 6000 Å is used and the n^{th} bright fringe is obtained at a point 'P' on the screen. Keeping the same setting, the source of light is replaced by green light of wavelength 5000 Å and now $(n+1)^{\text{th}}$ bright fringe is obtained at the point P on the screen. The value of 'n' is

1) 4

2) 5

3) 6

4) 3

Ans: 2**Sol :** When bright fringes are coinciding at a point on screen

$$n_1 \beta_1 = n_2 \beta_2$$

$$n_1 \frac{D}{d} \lambda_1 = n_2 \frac{D}{d} \lambda_2$$

$$n_1 \lambda_1 = n_2 \lambda_2$$

$$n \lambda_{\text{longer}} = (n+1) \lambda_{\text{shorter}}$$

$$n \times 6000 = (n+1) 5000$$

$$6n = 5n + 5$$

$$n = 5$$

Physics - Question 104

Original paper page 39; original question and option regions

104. Two charges each of charge $+10\ \mu\text{C}$ are kept on Y-axis $y=-a$ and $y=+a$ respectively. Another point charge $-20\ \mu\text{C}$ is placed at the origin and given a small displacement x ($x \ll a$) along X-

axis. The force acting on the point charge is $\left(x \text{ and } a \text{ are in metres, } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right)$

1) $\frac{3.6x}{a^2} N$

2) $\frac{2.4x^2}{a} N$

3) $\frac{3.6x}{a^3} N$

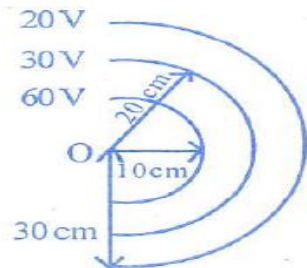
4) $\frac{4.8x}{a^2} N$

Ans : 3

Physics - Question 106

Original paper page 41; original question and option regions

106. The figure shows equipotential surfaces concentric at 'O'. The magnitude of electric field at a distance ' r ' meters from 'O' is



1) $\frac{9}{r^2} \text{Vm}^{-1}$

2) $\frac{16}{r^2} \text{Vm}^{-1}$

3) $\frac{2}{r^2} \text{Vm}^{-1}$

4) $\frac{6}{r^2} \text{Vm}^{-1}$

Ans : 4

Physics - Question 107

Original paper question page 42; original question and option regions

107. A region contains a uniform electric field $E = (10\hat{i} + 30\hat{j}) \text{ Vm}^{-1}$. A and B are two points in the field at (1, 2, 0) m and (2, 1, 3) m respectively. The work done when a charge of 0.8 C moves from A to B in a parabolic path is

1) 8 J

2) 80 J

3) 40 J

4) 16 J

Ans : 4

$$\text{Sol : } \vec{E} = (10\hat{i} + 30\hat{j}) \text{ Vm}^{-1}; q = 0.8 \text{ C}$$

$$\vec{r}_1 = \hat{i} + 2\hat{j}; \vec{r}_2 = 2\hat{i} + \hat{j} + 3\hat{k}$$

$$\vec{r} = \vec{r}_2 - \vec{r}_1 = \hat{i} - \hat{j} + 3\hat{k}$$

$$w = \vec{F} \cdot \vec{r} = \vec{E} q \cdot \vec{r}$$

$$= q \vec{E} \cdot \vec{r}$$

$$= 0.8 (10\hat{i} + 30\hat{j}) \cdot (\hat{i} - \hat{j} + 3\hat{k})$$

$$= 0.8(10 - 30) = 0.8 \times -20$$

$$= -16 \text{ J}$$

$$w = 16 \text{ J}$$

Physics - Question 108

Original paper page 42; original question and option regions

108. When a long straight uniform rod is connected across an ideal cell, the drift velocity of electrons in it is v . If a uniform hole is made along the axis of the rod and the same battery is used, then the drift velocity of electrons becomes

1) v

2) $>v$

3) $<v$

4) zero

Ans : 1

Sol : $i = NAeV_d$

$$\text{But } i = \frac{E}{R}$$

E is emf of cell

Physics - Question 109

Original paper page 43; original question and option regions

109. In a meter bridge experiment, when a nichrome wire is in the right gap, the balancing length is 60 cm. When the nichrome wire is uniformly stretched to increase its length by 20% and again connected in the right gap, the new balancing length is nearly

- 1) 61 cm 2) 31 cm 3) 51 cm 4) 41 cm

Ans : 3

$$\text{Sol : } \frac{R_{\text{left}}}{R_{\text{right}}} = \frac{l}{100 - l}$$

$$\frac{R_l}{R_r} = \frac{60}{100 - 60} \Rightarrow \frac{R_l}{R_r} = \frac{3}{2}$$

When wire is stretched

Volume before stretch = volume after stretching

$$A_1 l_1 = A_2 l_2 \Rightarrow \frac{A_1}{A_2} = \frac{l_2}{l_1}$$

$$R = \frac{\rho l}{A}; R \propto \frac{l}{A}$$

$$\frac{R_1}{R_2} = \frac{l_1}{l_2} \frac{A_2}{A_1}; \frac{R_1}{R_2} = \frac{l_1}{l_2} \frac{l_1}{l_2}$$

$$\frac{R_1}{R_2} = \left(\frac{l_1}{l_2} \right)^2$$

$$l_2 = l_1 + \frac{20}{100} l_1 = \frac{120}{100} l_1 = \frac{6}{5} l_1$$

$$\therefore \frac{R_1}{R_2} = \left(\frac{5l_1}{6l_1} \right)^2$$

$$\frac{R_1}{R_2} = \frac{25}{36} \Rightarrow R_2 = \frac{36}{25} R_1$$

Physics - Question 110

Original paper page 44; original question and option regions

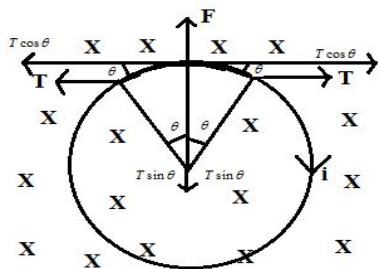
110. A loop of flexible conducting wire lies in a magnetic field of 2.0 T with its plane perpendicular to the field. The length of the wire is 1 m. When a current of 1.1 A is passed through the loop, it opens into a circle, then the tension developed in the wire is

1) 0.15 N

2) 0.25 N

3) 0.35 N

4) 0.45 N

Ans : 3**Sol :** $B = 2\text{ T}$, $l = 1\text{ m}$, $i = 1.1\text{ A}$ 

Consider small part of circle of length dl making an angle 2θ at the centre

Force due to magnetic field on the small element of $= F = Bidl$

This force is balanced by $2T \sin \theta$

$$2T \sin \theta = Bidl$$

when θ is small $\sin \theta = \theta$

$$2T\theta = Bi(R2\theta) \Rightarrow T = BiR$$

$$\text{But } l = 2\pi R \Rightarrow R = \frac{l}{2\pi}; T = Bi \frac{l}{2\pi} = 2 \times 1.1 \times \frac{1}{2 \times \frac{22}{7}}$$

$$T = 0.35\text{ N}$$

Physics - Question 111

Original paper page 45; original question and option regions

111. A charge q is spread uniformly over an isolated ring of radius ' R '. The ring is rotated about its natural axis with an angular velocity ' ω '. Magnetic dipole moment of the ring is

1) $\frac{q\omega R^2}{2}$

2) $\frac{q\omega R}{2}$

3) $q\omega R^2$

4) $\frac{q\omega}{2R}$

Ans : 1**Sol :** Due to rotation of uniformly charged ring electric current is i

$$i = \frac{q}{T} \text{ where } T \text{ is time period}$$

$$i = \frac{q}{\left(\frac{2\pi}{\omega}\right)} \Rightarrow i = \frac{q\omega}{2\pi}$$

Magnetic dipole moment of a ring = iA

$$= \frac{q\omega}{2\pi} \times \pi R^2$$

$$= \frac{1}{2} q\omega R^2$$

Physics - Question 112

Original paper page 45; original question and option regions

112. A magnetic dipole of moment 2.5 Am^2 is free to rotate about a vertical axis passing through its centre. It is released from East-West direction. Its kinetic energy at the moment it takes North - South position is $(B_H = 3 \times 10^{-5} \text{ T})$.

1) $50 \mu\text{J}$

2) $100 \mu\text{J}$

3) $175 \mu\text{J}$

4) $75 \mu\text{J}$

Ans : 4

Sol : $M = 2.5 \text{ Am}^2$

Work done = change in KE = $KE_{\text{final}} - KE_{\text{initial}} = KE_{\text{final}}$

But work done = $MB_H [\cos \theta_1 - \cos \theta_2]$

$= MB_H [\cos 90 - \cos 0]$

$= -MB_H$

$= 2.5 \times 3 \times 10^{-5}$

$= 7.5 \times 10^{-5} \text{ J}$

$= 75 \times 10^{-6} \text{ J} = 75 \mu\text{J}$

$KE_{\text{final}} = 75 \mu\text{J}$

Physics - Question 113

Original paper page 45; original question and option regions

113. A branch of a circuit is shown in the figure. If current is decreasing at the rate of 10^3 A s^{-1} , the potential difference between A and B is



1) 1 V

2) 5 V

3) 10 V

4) 2 V

Ans : 1

Physics - Question 114

Original paper page 46; original question and option regions

114. The natural frequency of an LC circuit is 125 kHz. When the capacitor is totally filled with a dielectric material, the natural frequency decreases by 25 kHz. Dielectric constant of the material is nearly

1) 3.33

2) 2.12

3) 1.56

4) 1.91

Ans : 3

Sol : $\omega = \frac{1}{\sqrt{LC}}$ $f = \frac{1}{2\pi\sqrt{LC}}$

$$f_1 = \frac{1}{2\pi\sqrt{LC}}; f_2 = \frac{1}{2\pi\sqrt{L(KC)}} \qquad \frac{f_2}{f_1} = \frac{1}{\sqrt{k}}$$

$$K = \left(\frac{f_1}{f_2} \right)^2$$

$$\therefore K = \left(\frac{125}{100} \right)^2 = \left(\frac{5}{4} \right)^2 = \frac{25}{16} = 1.56$$

Physics - Question 115

Original paper page 46; original question and option regions

115. Choose the correct sequence of the radiation sources in increasing order of the wavelength of electromagnetic waves produced by them.

- 1) X-ray tube, Magnetron valve, Radio active source, Sodium lamp
- 2) Radio active source, X-ray tube, Sodium lamp, Magnetron valve
- 3) X-ray tube, Magnetron valve, Sodium lamp, Radio active source
- 4) Magnetron valve, Sodium lamp, X-ray tube, Radio active source

Ans : 2

Sol : Conceptual

Physics - Question 116

Original paper page 46; original question and option regions

116. A photo sensitive metallic surface emits electrons when X-ray of wavelength ' λ ' fall on it. The de Broglie wavelength of the emitted electrons is (Neglect the work function of the surface, m is mass of the electron. h-Planck's constant, c-velocity of light)

1) $\sqrt{\frac{2mc}{h\lambda}}$

2) $\sqrt{\frac{h\lambda}{2mc}}$

3) $\sqrt{\frac{mc}{h\lambda}}$

4) $\sqrt{\frac{h\lambda}{mc}}$

Ans : 2**Sol :** Energy of X-rays = energy of electron

$$\frac{1}{2}mv^2 = \frac{hc}{\lambda}$$

$$\text{or } \frac{p^2}{2m} = \frac{hc}{\lambda}$$

$$p = \sqrt{\frac{2mhc}{\lambda}}$$

De - broglie wavelength of electron

$$= \lambda_e = \frac{h}{p} = \frac{h\sqrt{\lambda}}{\sqrt{2mhc}} = \sqrt{\frac{h\lambda}{2mc}}$$

Physics - Question 117

Original paper page 47; original question and option regions

117. An electron in a hydrogen atom undergoes a transition from a higher energy level to a lower energy level. The incorrect statement of the following is

- 1) Kinetic energy of the electron increases
- 2) Velocity of the electron increases
- 3) Angular momentum of the electron remains constant
- 4) Wavelength of de-Broglie wave associated with the motion of electron decreases

Ans : 3

Sol : KE of electron in orbit of radius r

$$= \frac{e^2}{8\pi \epsilon_0 r}$$

(1) when r decreases, KE increases

$$(2) KE = \frac{1}{2}mv^2$$

KE increases $\Rightarrow v$ increases

$$(3) L = \frac{nh}{2\pi} \quad n \text{ decreases} \Rightarrow L \text{ decreases}$$

$$(4) n\lambda = 2\pi r$$

$$r \propto n^2$$

$$\Rightarrow \lambda \propto n \quad \therefore \lambda \text{ decreases}$$

In correct option is (3)

Physics - Question 118

Original paper page 47; original question and option regions

118. The radius of germanium (Ge) nuclide is measured to be twice the radius of ${}^9_4\text{Be}$. The number of nucleons in Ge will be

1) 72

2) 73

3) 74

4) 75

Ans : 1

Sol : $R = R_0 A^{1/3}$

$$\therefore R \propto A^{1/3}$$

$$\Rightarrow A \propto R^3$$

$$\frac{A_2}{A_1} = \frac{R_2^3}{R_1^3}$$

$$\frac{R_2}{R_1} = 2$$

$$\therefore \frac{A_2}{A_1} = 8 \quad \therefore A_2 = 8 A_1 = 8 \times 9 = 72$$

Physics - Question 119

Original paper page 47; original question and option regions

119. For a common-emitter transistor amplifier, the current gain is 60. If the emitter current is 6.6 mA then its base current is

1) 6.492 mA

2) 0.108 mA

3) 4.208 mA

4) 0.343 mA

Ans : 2

Sol : Current gain $\beta = 60 = \frac{I_C}{I_B}$

Physics - Question 120

Original paper page 48; original question and option regions

120. If a transmitting antenna of height 105m is placed on hill, then its coverage area is

1) 4224 km^2

2) 3264 km^2

3) 6400 km^2

4) 4864 km^2

Ans : 1

Sol : $d = \sqrt{2Rh}$

$$d = \sqrt{2 \times 6400 \times 1000 \times 105}$$

$$d = \sqrt{1344 \times 10^6}$$

$$\text{Area} = \pi d^2 = \pi \times 1344 \times 10^6 \text{ m}^2$$

$$= 4224 \text{ km}^2$$

CHEMISTRY